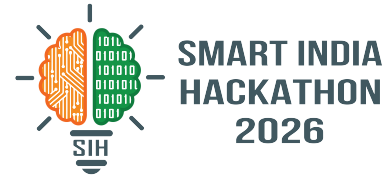
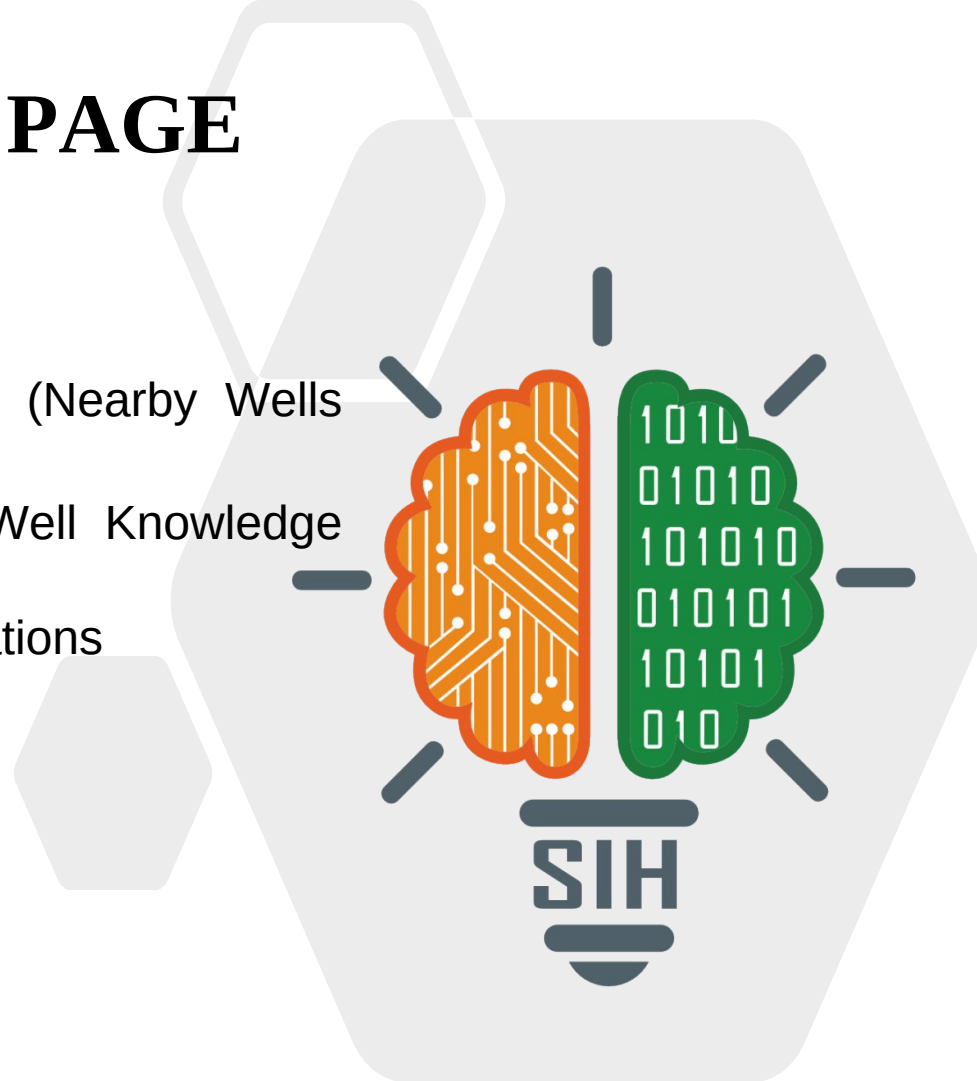


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26121
- **Problem Statement Title-** eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 170820
- **Team Name -** Pyrophorics_787057



IDEA TITLE

AI-Powered Well Intelligence & Drilling Support System

Problem

Historical drilling knowledge from nearby wells is **trapped in unstructured PDFs**, daily reports, and **scattered databases**.

Drilling engineers **lack real-time, context-aware access to past hazards** (e.g., stuck pipe, mud loss, NPT etc.) during active drilling.

Manual review of offset data is **time-consuming**, leading to delayed decisions and increased Non-Productive Time.

Detailed Explanation of Our Idea

Privacy Friendly System: Uses local Vision AI (Qwen2.5-VL) to digitize sensitive old documents into clean Markdown.

Live Synchronization: During live operations, it constantly correlates real-time telemetry with historical incidents from nearby wells.

Knowledge Accessibility: Added a searchable knowledge repository of drilling events, lessons learned, operational challenges, and mitigation measures.

Our Idea

An AI-powered system that bridges the gap between **historical well data** and **real-time drilling operations**.

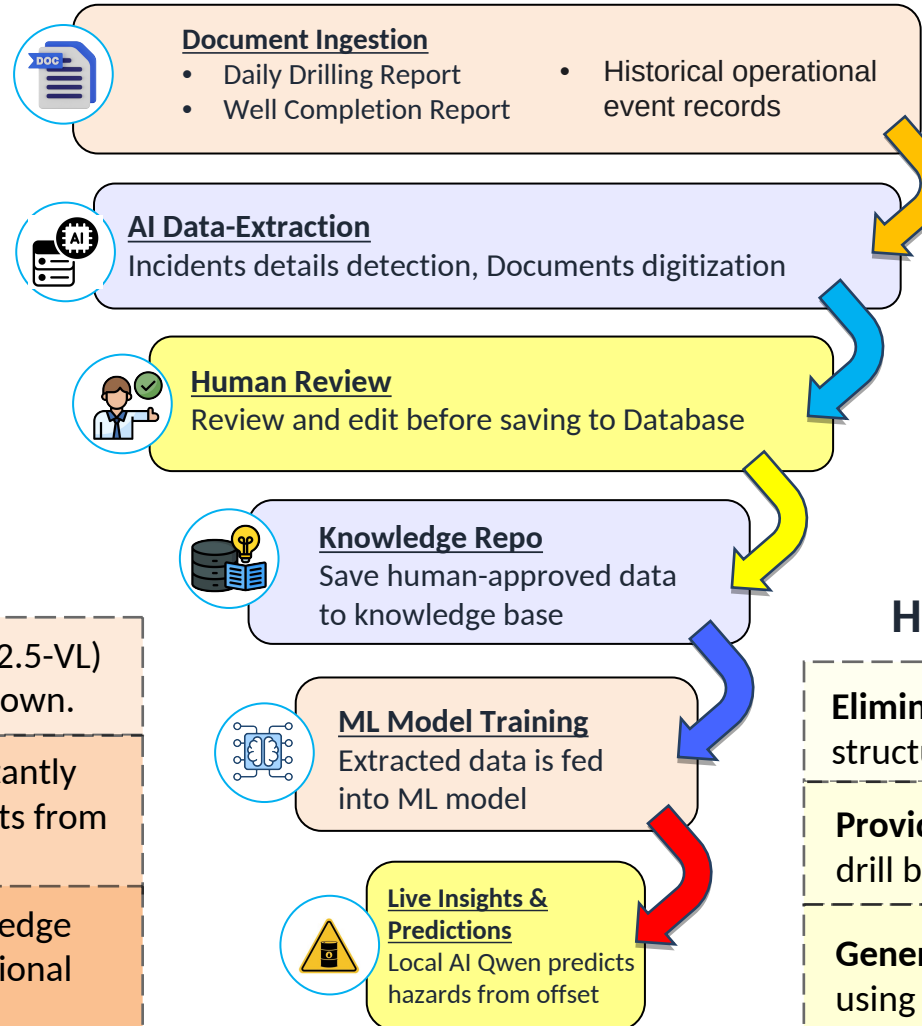
It **ingests multi-page documents**, **extracts information** using advanced Vision LLMs, and maps historical risks to the active well's current depth.

How it Addresses the Problem

Eliminates manual document searching by structuring all offset well history.

Provides proactive visual alerts right before the drill bit reaches a historically hazardous zone.

Generates instant prescriptive mitigations using an AI Copilot when a live risk is detected.



TECHNICAL APPROACH

Tech Stack

Frontend

React & Vite: High-performance, single page application dashboard.
Tailwind CSS & Recharts: Responsive UI and real-time data visualizations.
React-Leaflet: Interactive geospatial mapping for offset well distance calculations.

Backend

Node.js & Express.js: Handles REST APIs, document uploads, and business logic.
WebSockets (Socket.io): Streams 1-Hz live drilling telemetry to the dashboard.
Security: JWT authentication, bcrypt password hashing, and role-based access.

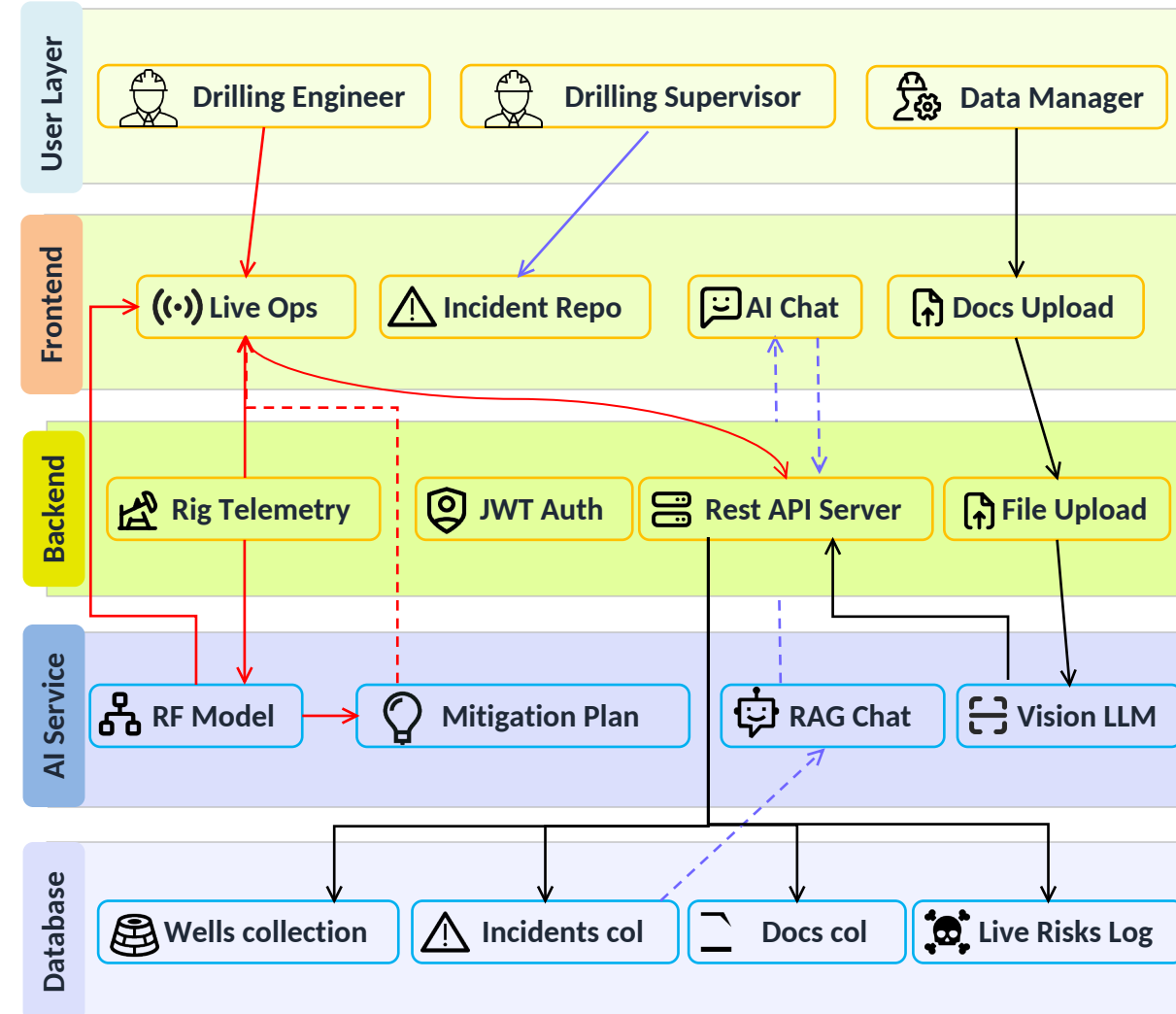
Database

MongoDB: Stores structured well metadata, hashed documents, and extracted hazard events.
Schema Design: Optimized schemas capable of linking extracted markdown pages directly to their source file ID.

AI

PyMuPDF (fitz): Ultra-fast, dependency-free native rendering of PDF pages into images.
Vision LLMs: Uses local Qwen2.5-VL in Ollama to perform complex OCR, diagram description, and Markdown formatting.
Prediction: Uses same Qwen for risk alerts and making predictions from offset data.

System Architecture



Technical Feasibility

- Built entirely on modern, open-source tech stacks (React, Node, FastAPI).
- AI services work even if there's no internet connectivity in remote locations

Financial Feasibility

- **No AI licensing costs:** The AI model used is free and open-source, requiring no subscription.
- **No expensive hardware:** Runs on even a 4GB RAM Laptop, no specialized GPU infrastructure required.

Operational Feasibility

- **Easy to adopt:** If an engineer can use a standard dashboard on a browser, they can use this system, no special training required.
- **No trust issues with AI:** Every AI output is reviewed and approved by a human engineer before it is saved. The system never makes autonomous decisions.

Challenges & How We Overcome



Data Security & Privacy: Drilling data is highly confidential, sending it to external AI servers is a security risk



The system **supports fully offline mode** where the AI runs locally on the company's own intranet server. Sensitive well data never leaves the organization's network.



Safety: Wrong AI prediction in critical moments



AI outputs are advisory only; operator must confirm before any actuator action



Legacy drilling **reports are decades old, poorly scanned, or handwritten**, standard text tools fail on them.



The system uses **Vision AI (Qwen2.5-VL)** that treats each page as an image and reads it the same way a human would, regardless of scan quality, formatting, or layout. No clean digital text is required.

IMPACT AND BENEFITS

Impact on Target Users

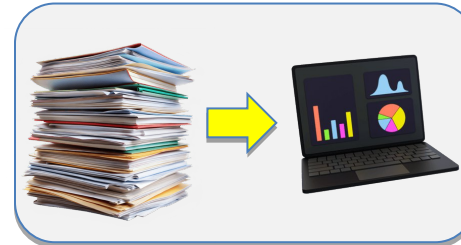
- **Drilling Engineers:** Instantly receive historical context and AI-generated mitigation plans exactly when they approach a dangerous depth.
- **Data Managers:** Effortlessly digitize decades of legacy PDFs without manual data entry.

Impact on Organisation (OIL / ONGC)

- Creates a centralized, searchable drilling knowledge base that grows with every new well drilled, turning isolated field experience into a permanent organizational asset.

Builds Institutional Knowledge

- The system preserves the hard-won experience of senior engineers as structured, searchable data that future teams can benefit from, even after those engineers retire.



Economic Benefit

- **Reduces Non-Productive Time (NPT):** Every hour of NPT in drilling costs lakhs of rupees. By proactively alerting engineers before hazards are hit, the system directly cuts costly downtime from stuck pipes, mud losses, and kicks.

Environmental Benefit

- **Prevents blowouts and uncontrolled kicks:** Early hazard detection reduces the risk of catastrophic well control incidents that can cause oil spills and long-term ecological damage to surrounding land and water bodies.

Social Benefit

- **Protects lives on the rig:** Proactive alerts for hazards like Overpressure Kicks and Stuck Pipe directly reduce the risk of accidents that can injure or kill rig workers.

IMPACTS
BENEFITS

RESEARCH AND REFERENCES

Chandrasekaran, S. et al. (IIT Madras): "Machine Learning Applications in Petroleum Engineering: A Review"

[Link](#)

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Ministry of Petroleum & Natural Gas, Government of India: "Hydrocarbon Exploration and Licensing Policy (HELP) & Digital India in Energy Sector"

[Link](#)

Comptroller and Auditor General (CAG) of India: "Performance Audit: Hydrocarbon Exploration Efforts of Oil India Limited"

[Link](#)

ONGC Corporate Communications: "Real-Time Operations Centre (RTOC) Phase-II, Institute of Drilling & Well Engineering, Dehradun"

[Link](#)

ONGC: "Well Information System (WIS): Generative AI for Daily Drilling Reporting across Offshore Rigs"

[Link](#)

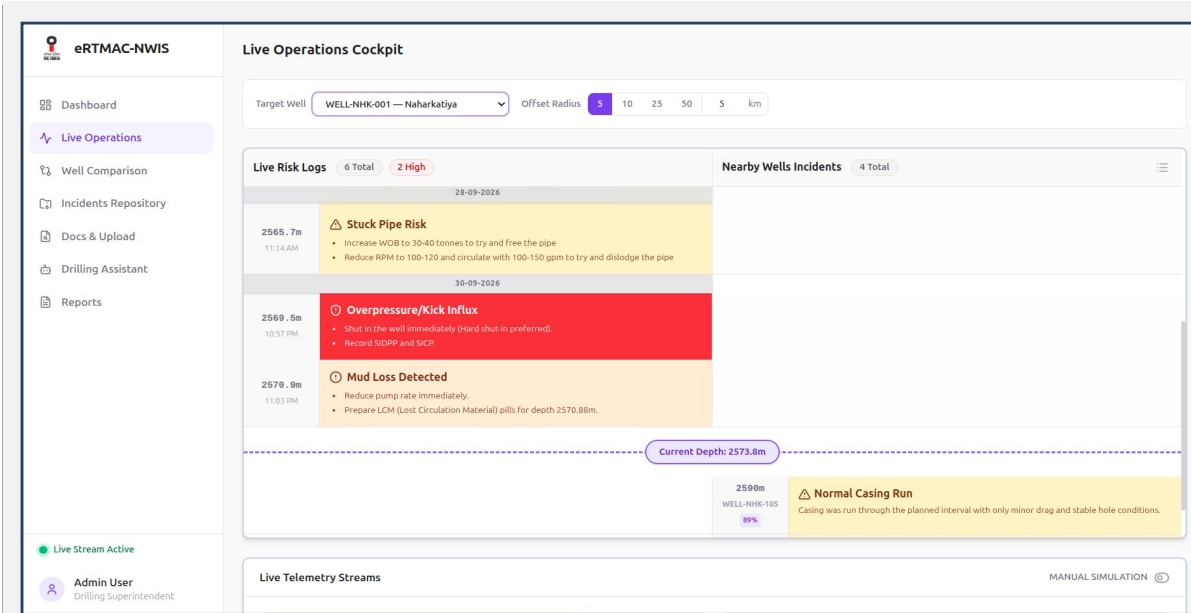
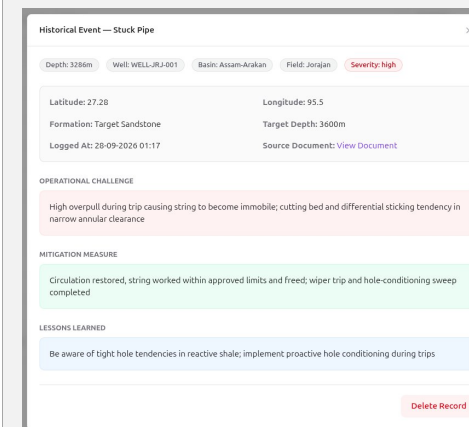
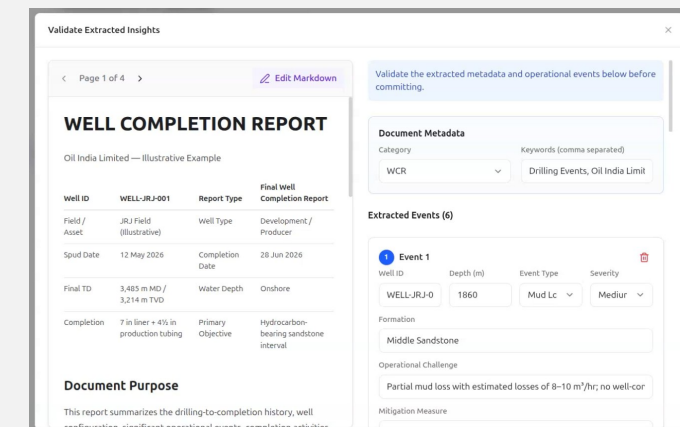


Fig: NWIS UI — Left: Live Risk Logs, Right: Nearby Wells Past Incidents



Past Incident Details



Data Extraction from PDFs